Problem Statement and Goals
Software Engineering

Team #14, Troitsky Toolbox
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Table 1: Revision History

Date	Developer(s)	Change
September 22th, 2026	Pedro, Mason, Jack, Michael	Discussion and notes of document contents
September 24th, 2026	Pedro	Initial draft problem statement section from discussion
Septemeber 26th, 2026	Pedro	Goals and reflection section
Septemeber 27th, 2026	Mason	Review and grammar adjustments
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1 Problem Statement

1.1 Problem

The McMaster Troitsky club is a student-run organization that registers and supports 4-6 teams of students as they take on the task of modelling, analyzing and manufacturing a bridge out of popsicle sticks, white glue and dental floss. While the club has been successful in the past, results often vary in performance both between years and teams, due to the lack of centralized information and technical support for members. As teams progress through several years of competition, undergoing material testing, bridge design and analysis, and even experimenting with more innovative concepts, skill and knowledge within

the club often consolidates within more experienced members or single teams, ensuring that when those members graduate, information is lost to the club. This project seeks to create a centralized information repository and software subteam providing technical support for the McMaster Troitsky club, ensuring information transfer between years and teams, and facilitating support and onboarding for new members.

1.2 Inputs and Outputs

- **Inputs:** Range of McMaster Troitsky Club members' and alumni knowledge to build the repository. Past documentation of the club and team specific activities, along with their collected data, will be input into the suitable repositories within the service.
- **Outputs:** Toolbox of services to support the McMaster Troitsky club in information centralization and technical support for members. A simplified process for searching and accessing the range of information the team has collected and will continue to collect.

1.3 Stakeholders

1.3.1 Primary Stakeholders

- **McMaster Troitsky Club General Members**
  The general members of the club are primary stakeholders as they will be the main users and those that we will focus on supporting with our project. These users will be both supporting and expanding the data the system provides, and will be the main beneficiaries of the system.
- **McMaster Troitsky Club Executive Team**
  The McMaster Troitsky Club Executive Team are primary stakeholders as they will benefit from both the organizational support the service provides, and the alleviation of the burden of technical support and information transfer that is currently placed on them. Their input will be essential for service design and prioritization of features.

1.3.2 Secondary Stakeholders

- **Concordia University's Troitsky Organization**
  As Concordia University hosts the Troitsky competition, they establish the rules and regulations of the competition, and therefore changes what is prioritized by the primary stakeholders. This would influence the design of the system, and what features would benefit teams the most.
- **MES - McMaster Engineering Society**
  The MES organizes engineering clubs and seeks to benefit from the legitimization of the McMaster Troitsky Club through a centralized information platform. The MES also provides funding for the McMaster Troitsky

Club, so the hosting of services and associated costs may be dependent on the given budget.

1.3.3 Tertiary Stakeholders

- **McMaster University**
  The university itself is a tertiary stakeholder as it benefits from the success of the McMaster Troitsky Club, and the success of teams during the Troitsky competition.

1.4 Environment

1.4.1 Software Environment

The Troitsky Toolbox will be a web-based application, accessible through any common modern web browser, namely Chrome, Edge, Firefox and Safari. A landing for the club will be generally accessible, with services and stored data limited to authorized members.

2 Goals

2.1 Information Centralization and Access

Create a centralized information repository, allowing users to browse past designs, material testing information, and other relevant development information. Users should be able to search and filter through information, locating relevant content quickly and efficiently. The system should allow for the addition of new information, streamlining the process of growing the repository.

2.2 Key Model Analysis Features and Bridge Predictions

Provide a platform for teams to upload their bridge models through a standardized format, allowing the user to view and analyze the model, highlighting key metrics and predictions about the bridge's performance. This should decrease the barrier to entry to bridge analysis, allowing all members of a team to get involved in the analysis process, even without formal modelling knowledge. This will be integrated with the material testing support to provide more accurate bridge analysis from the stored material properties.

2.3 Material Testing Support and Facilitated Result Analysis

As the McMaster Troitsky club begins to standardize knowledge collection, material testing data will be collected and stored in a standardized format. The service will streamline this process, allowing testing results to be easily uploaded and analyzed against past results, along with collecting supplementary testing information for future years.

2.4 Simplified and Comprehensive Knowledge Transfer

Support documentation from the McMaster Troitsky club through exit documents and other knowledge transfer methods, ensuring that information and designs are consolidated and accessible to future members. Specialized member roles, such as executive positions, will also be able to keep a record of their role responsibilities and key factors, to further improve the club organization.

2.5 Integration of Software Development into the Club

The maintenance and consistent development of the Troitsky Toolbox will be integrated into the club operations, creating a subteam around software operations that will recruit interested, relevant members. The operation of the system rests on its ability to be maintained once the team is no longer handling anything, so this goal is fundamentally crucial to the success of the project. This will be assessed through hand-off and exit documents that are left outlining what roles on the subteam entail.

3 Stretch Goals

3.1 Preparing for Future Development, Maintenance, and Expansion of Project

In the process of designing this toolbox, the services will be created with expansion heavily considered. While during initial development of services such as the model analysis may be simpler, they will be developed with room for a specialized model to be implemented in the future. The project will utilize heavily structured code and design patterns, along with thorough documentation to create a solid platform for future updates.

4 Custom Documents (CDs)

4.1 Interviews

Interviews will be the first form of CD performed, used to gather a large swath of specified and verifiable valuable user opinion data. With interviews, more experienced members can have the space to inform the team with as much domain knowledge about what the project requires as they are able to, helping to specialize the output to true user needs.

4.2 Usability Testing

Due to the nature of this service, the user experience will be vital to the project's success. During the second term, the application should be functional and already providing some value to the users, so usability testing will provide maximal value allowing the team to assess how the services come together as a whole.

Appendix

Reflection

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

Jack Easton

1. During writing the this deliverable, it really gave our team a great opportunity to come together and unify our perpsectives and ideas. All team members had many invaluable ideas and inputs on the project which allowed us to end up with a proposed development plan and problem statement that is greater than what we could have come up with individually.
2. Intially when writing the content and documentation for this deliverable, it was difficult to focus all of the different ideas, options, tools, and pieces of domain knowledge provided by the Troitsky Team into a clear and articulatable form. However, through working through proven processed for development and figuring things out one question/bulletpoint at a time, I feel much more confident in my understanding and ability to convey our plan for the POC document.
3. I think the key of the scope of our project being suitable for a Capstone project lies in the unique structure of our project. Because concept of the project is to create a toolbox of a variety of independent but specially customized tools, each tool presents isolated complexity. This naturally allows the project to be broken down into attainable sections (by tool). When viewed individually, some of the tools are highly complex (making them suitable for a senior design project), but all of the tools do not relpy on each other to present their functionality. It is additionally worth noting that a large part of the complexity of the project comes in how to correctly and efficently present a multitude of differing features while adhering to proper software architecture design principles.

Pedro Azevedo

1. During the writing process of this deliverable, I gained an understanding of the core needs of the club, and therefore the primary goals that bring value to this club. As I have been a part of the club for the last 4 years, the domain knowledge of the club operations and principal pain points was already known. The selection of primary stakeholders also went well

as the team is well acquainted with clubs at the school, so understanding who would be affected was straightforward. This also simplified the first choice of CD to be interviews, as those would be easily completed as I have direct access to organizing with the club.

2. The main pain points during this deliverable came from the initial organization of the project, and the learning curve associated with the heavy use of issues, projects and milestones. Once that was set up, the distribution of work became straightforward, along with the subsequent review process that each ticket must go through.

3. As this project's structure is more a collection of services, the scope of the project will be more adjustable throughout the year, with certain services being prioritized and others seen more as a bonus feature. At the beginning, the team considered focusing on specialized numerical modelling for the bridge simulation, but felt that the value we could produce lay more in the variety of services described in this document.

Mason Azzopardi

1. We had a very strong idea of who our stakeholders were before this deliverable, so to narrow down the problems we wanted to tackle and how we wanted to tackle them was just a series of brainstorming sessions; it always felt like we had a clear path of what we had to do next.

2. One large pain point was the integration of the Custom Documents, they aren't included within the given rubric or any examples I could find, the short description of how to define them left us unsure if we defined them with enough detail. It left us in a state of not wanting to overly narrow the scope of our CDs while still putting down enough information for it to be helpful in this document.

3. The scope of the goals was refined iteratively, first through a few conversations with the Troitsky executives and our team members where a custom physics modelling system was considered. Then through a conversation with Dr Spencer Smith we settled on a more realistic idea to have a toolbox of subservices that Troitsky is missing. The exact subservices and thier scopes were outline in a few team meetings after our coversation with Dr. Smith and wil continue to be adjusted throughout the course of this project.

Michael Roberts

1. We maintained an open forum for discussing ideas regarding this project, and after oral agreement, transferring it into writing was seamless.

2. One large pain point was narrowing the grand ambitions we had into a managable list of goals. Another was adjusting to GitLab in general, especially finding the rubrics used for grading.

3. Our goals were modified through an iterative process, by reseraching resources currently available to draw on, consulting with the Troitsky club, our stakeholders, and abandoning a large modelling application which has a significant chance of non-completion even after the 8 months of capstone.

Generative AI Use Disclosure

No AI was used in the creation of this document, except for the spell-checking and grammar services built into the Visual Studio Code editor. Some common free models were used to guide research into tool choice and development plan choices, specifically the core libraries and frameworks adopted for the project.